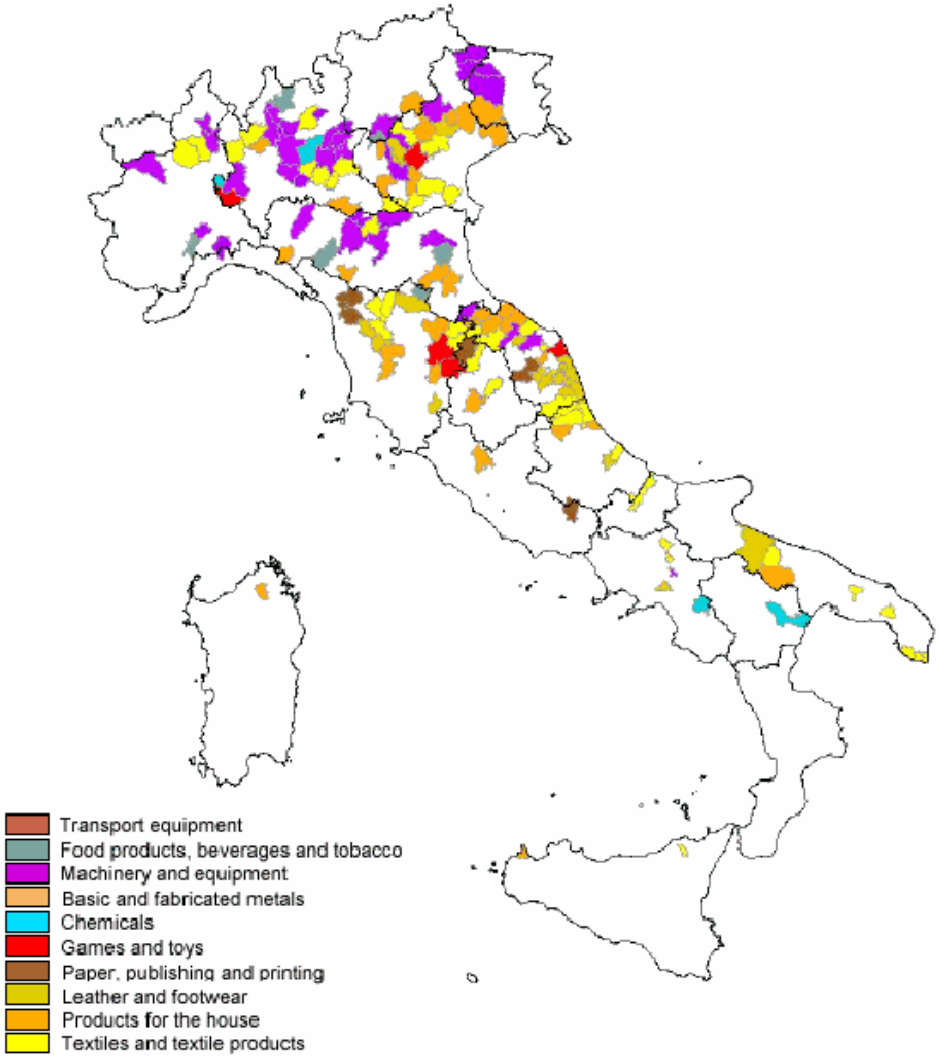
Labor markets and skills: labor mobility, skill-relatedness

Zoltán Elekes

Economic Complexity, Class 5 – Oct 14 2026

Social Data Science, Corvinus University of Budapest

Specialization versus diversity



Italian industrial districts

- SMEs (Small and Medium-sized Enterprises)
- Local traditions and strong cohesion
- Craft industry heritage
- Small town and its surroundings
- Business development organizations
- Supportive local governments

Specialization: export base

Location Quotient

•Indicates the **degree of specialization** of a region in a given industry.

•Compares the **local share of employment** in the industry to the **national average**.

•Assumption: the **specialized product of the region is not only consumed locally**, thus the region participates in **interregional trade**.

$$LQ = \frac{E_{ir}}{E_r} \Big/ \frac{E_{in}}{E_n}$$



Specialization: examples

Many firms of the same industry in one place

- **Detroit, USA:** automobiles – Ford, GM, Chrysler and thousands of parts suppliers
- **Silicon Valley, USA:** chips, later software – firms spun out of each other (Fairchild → Intel)
- **Sassuolo, Italy:** ceramic tiles – tile, kiln and glaze producers in one district
- **Győr and Kecskemét, Hungary:** car manufacturing – Audi, Mercedes-Benz and their suppliers

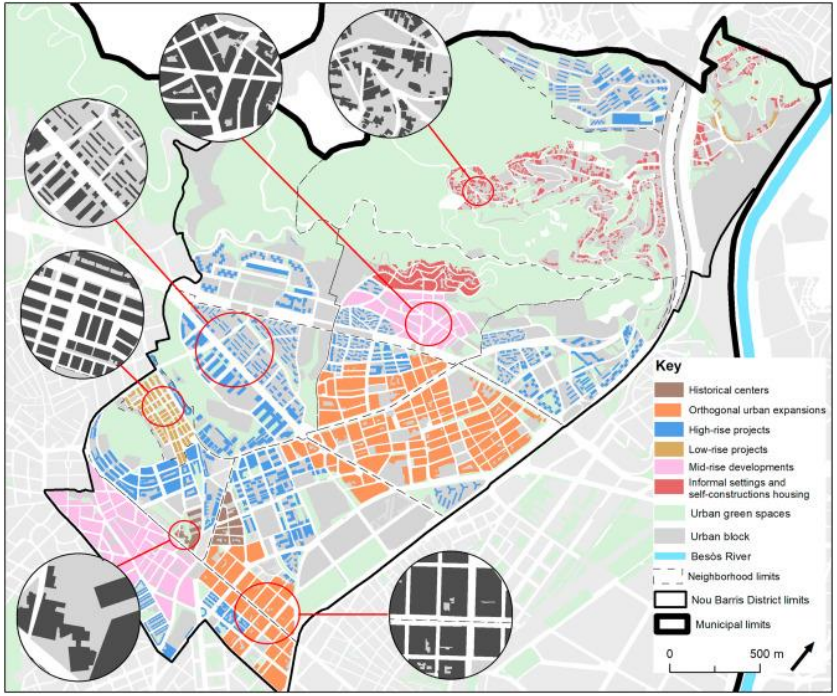
Why it may help growth (MAR, Porter)

- Shared suppliers and a pool of specialized workers
- Ideas spread **within** the industry: job-hopping, imitation, reverse engineering

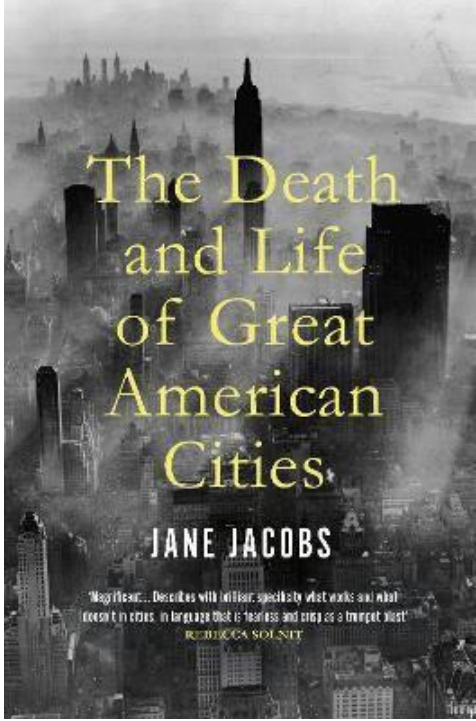
Diversity

Cities are arenas of interaction

Mixing in the built environment







Diversity: examples

Many different industries side by side

- **New York:** finance, media, fashion, advertising, tech, universities, health care
- **Budapest:** finance and business services, ICT, pharmaceuticals, creative industries, tourism

New industries are born where industries meet (Jacobs, 1969)

- **Dressmakers**, not lingerie makers, invented the bra
- **Grain and cotton merchants** in New York created financial services
- A **San Francisco food processor** short of capital invented equipment leasing

Examples from Jacobs (1969), as cited in Glaeser et al. (1992)

Glaeser et al. (1992): data

Unit of observation: the city-industry (e.g., Philadelphia apparel)

- 170 largest US cities, the 6 largest 2-digit industries in each (1956)
- 1,016 city-industries
- County Business Patterns 1956 and 1987: employment, payroll, establishments

Question

- Does the **same industry** grow faster in specialized or in diverse cities?

Outcome and controls

- Employment growth: log(employment 1987 / employment 1956)
- Controls: national growth of the industry, 1956 wage and employment, South

Glaeser, Kallal, Scheinkman & Shleifer (1992). Growth in Cities. Journal of Political Economy 100(6), Section III

Measuring the three externalities

Specialization
MAR, Porter: higher → faster growth

Competition
Porter, Jacobs: higher → faster growth

Diversity
Jacobs: lower share → faster growth

$$\text{specialization} = \frac{\text{industry employment in city/total employment in city}}{\text{industry employment in U.S./total employment in U.S.}}. \quad (8)$$

$$\text{competition} = \frac{\text{firms in city-industry/workers in city-industry}}{\text{firms in U.S. industry/workers in U.S. industry}}. \quad (9)$$

Finally, to address Jacobs's theory, we need a measure of a variety of industries in the city outside the industry in question. The measure we use is the fraction of the city's employment the largest five industries other than the industry in question account for in 1956. The mean of this ratio is .35: cities are not well diversified. The lower this ratio, the more diverse the city is and the faster the industry in question should grow according to Jacobs.

Glaeser, Kallal, Scheinkman & Shleifer (1992). Growth in Cities. Journal of Political Economy 100(6), eqs. (8)–(9) and p. 1142

The measures in the data

Fastest-growing city-industries

- lower concentration
- more competition
- more diverse cities (lower nondiversity)

Slowest-growing city-industries

- extreme concentration: Scranton anthracite mining 113× the US share
- weak competition

Also

- fast growers in the South and West, decliners in the East and Midwest

D. Industry Growth

City-Industry	Growth	Nondiversity	Competition	Concentration
	Five Fastest-Growing City-Industries*			
Albuquerque Business services	3.325	.217	1.500	1.090
San Jose, Calif. Electric machinery	2.765	.290	.835	2.582
San Jose, Calif. Durable wholesale trade	2.407	.310	1.008	.883
San Jose, Calif. Transportation equipment	2.403	.311	.930	.876
Atlantic City, N.J. Hotels	2.345	.373	.418	11.221
	Five Slowest-Growing City-Industries			
Scranton, Pa. Anthracite coal mining	−5.417	.387	.931	113.139
Manchester, N.H. Leather products	−5.161	.331	.272	19.559
Wilkes-Barre–Hazleton, Pa. Tobacco products	−5.078	.466	.279	21.193
Hamilton-Middletown, Ohio Primary metal industries	−4.813	.513	.326	4.271
Gadsden, Ala. Textile mills	−4.714	.406	.185	4.876

* Growth is log(employment in 1987/employment in 1956). Nondiversity is city's other top five industries' share of 1956 total city employment. Competition is establishments per employee relative to establishments per employee in the U.S. industry. Concentration is the city-industry's share of city employment relative to U.S. industry's share of U.S. employment in 1956.

Glaeser, Kallal, Scheinkman & Shleifer (1992). Growth in Cities. Journal of Political Economy 100(6), Table 1, Panel D

Results

Employment growth over 30 years

- **Specialization (–):** +1 SD → 12% **less** growth
- **Competition (+):** twice as many firms per worker as the US → +59%
- **Diversity (+):** other top-5 industries' share 0.1 lower → +9%

Verdict

- Negative on MAR, mixed on Porter, consistent with Jacobs

Caveat

- More firms per worker may just mean small firms grow faster

TABLE 3

CITY-INDUSTRY EMPLOYMENT GROWTH BETWEEN 1956 AND 1987

	LOG(EMPLOYMENT IN 1987/EMPLOYMENT IN 1956) IN THE CITY-INDUSTRY			
DEPENDENT VARIABLE	(1)	(2)	(3)	(4)
Constant	−.423	−.923	−.181	−.513
	(.129)	(.129)	(.159)	(.149)
Log(U.S. employment in 1987/U.S. employment in 1956) in the industry outside the city	1.140	1.209	1.237	1.148
	(.059)	(.052)	(.055)	(.056)
Wage in the city-industry in 1956	.0137	.0226	.0379	.027
	(.109)	(.104)	(.109)	(.104)
Employment in the city-industry in 1956 (in millions)	−2.898	−3.280	−3.91	−4.080
	(1.099)	(1.055)	(1.131)	(1.073)
Dummy variable indicating presence in the South	.426	.416	.370	.378
	(.057)	(.054)	(.058)	(.055)
City-industry's share of city employment relative to industry's share of U.S. employment in 1956	−.0128	. . .	. . .	−.00799
	(.003)			(.003)
Establishments per employee in the city-industry relative to establishments per employee in the U.S. industry in 1956	. . .	.587	. . .	.561
		(.057)		(.057)
City's other top five industries' share of total city employment in 1956	. . .	. . .	−.894	−.913
			(.259)	(.245)
Adjusted R^2	.392	.439	.387	.450
Number of observations	1,016	1,016	1,016	1,016

NOTE.—Standard errors of parameter estimates are in parentheses beneath these estimates.

Glaeser, Kallal, Scheinkman & Shleifer (1992). Growth in Cities. Journal of Political Economy 100(6), Table 3

From one paper to a debate

Open issue in Glaeser et al. → later research

- **Only mature industries** → Henderson, Kuncoro & Turner (1995): specialization helps mature industries; new high-tech industries also gain from diversity
- **Employment, not innovation** → Feldman & Audretsch (1999): diversity across industries sharing a science base drives innovation
- **Static snapshot** → Duranton & Puga (2001): "nursery cities" – products are born in diverse cities, then move to specialized ones
- **Broad 2-digit industries** → Frenken, Van Oort & Verburg (2007): related variety

Reviews: no universal winner

- Beaudry & Schiffauerova (2009); de Groot, Poot & Smit (2016): it depends on sector, life-cycle stage, outcome and measurement

Glaeser et al. (1992): three theories

	Spillovers come from	Local competition	Fastest growth in	Example in the paper
MAR (Marshall, Arrow, Romer)	within the industry	bad: monopoly lets innovators keep the returns	specialized cities, local monopoly	Silicon Valley chips
Porter (1990)	within the industry	good: rivalry forces innovation	specialized cities, fierce competition	Italian ceramics, gold jewelry
Jacobs (1969)	other industries	good: speeds up adoption of ideas	diverse cities, competition	the bra, invented by dressmakers

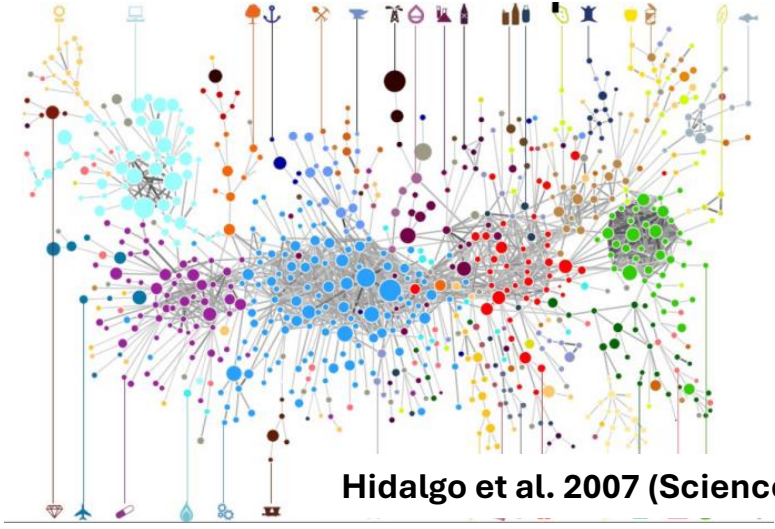
Glaeser, Kallal, Scheinkman & Shleifer (1992). Growth in Cities. Journal of Political Economy 100(6), pp. 1127–1132

Economic progress and urban success

- Social interaction: population density
- Learning in the city: industry structure
- Related knowledge is easier to learn but contains less novelty







Hidalgo et al. 2007 (Science)

Labour flows in evolving regional economies

Zoltán Elekes

KRTK KTI Közgazdaságtudományi Intézet
ANETILAB
UMEÅ UNIVERSITET
Regional Economies in Change

Capability-based view of economic development

Tools



Codes



Skills

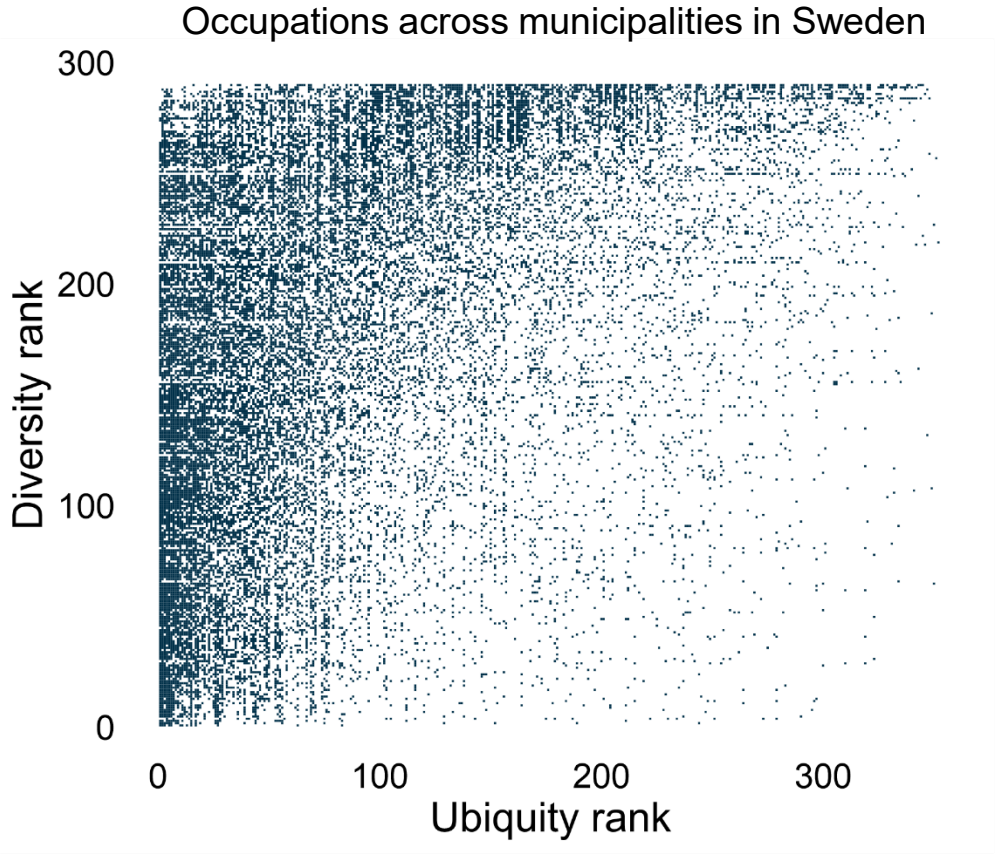


- Economic development entails the gradual accumulation of **complementary capabilities** to unlock more **sophisticated activities** (Hidalgo & Hausmann, 2009 PNAS; Stojkoski et al., 2023 ComEE).
- Resource-based view of the region: sustained competitive advantage comes from **valuable**, **rare**, **imperfectly imitable** and **non-substitutable** resources (Neffke et al., 2018 EG; Frenken et al., 2023 CJRES).

Regional economies are specialised and coherent

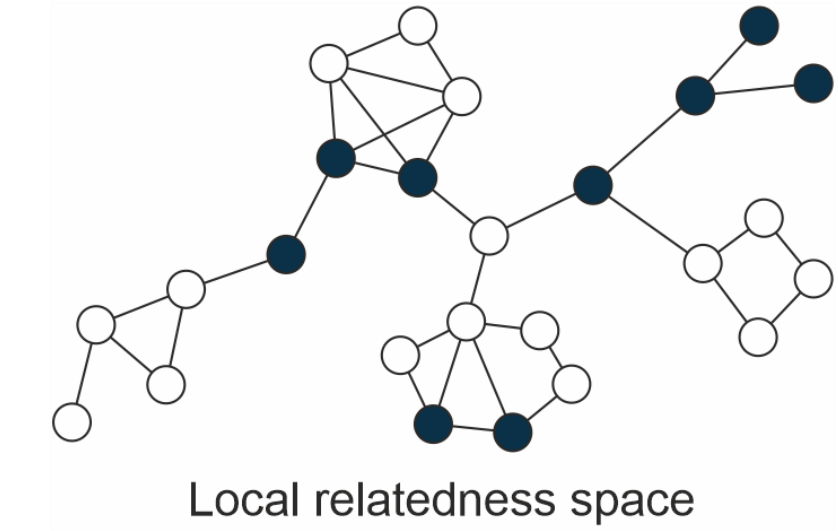


Economic activities cluster in space



- Many places tend to have relatively few specialisations while a few places have relatively many (e.g., Hidalgo et al., 2007 Science).
- Activity pairs tend to **co-agglomerate** (Ellison et al., 2010 AER; Diodato et al., 2018 JoUE; Juhász et al., 2026 JoEG):
  - Specialised labour pooling.
  - Specialised supply-chains.
  - Knowledge spillovers.

Relatedness spaces

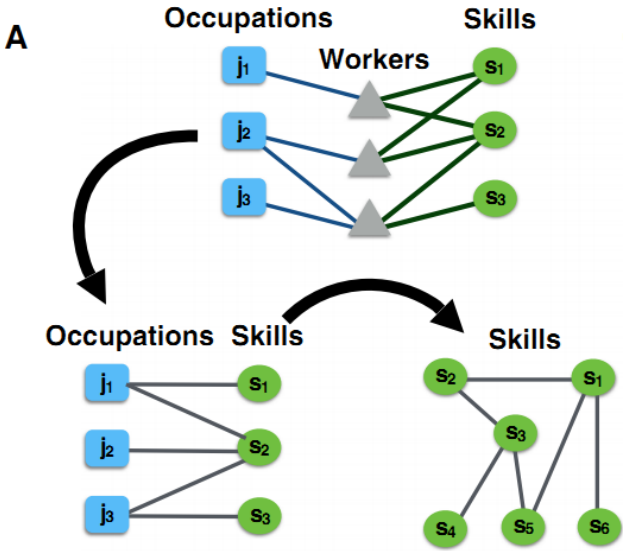


- Economic **networks** reflect a **division of labour** between the elements of an economy, and links reflect **solutions** to particular **co-ordination problems** inherent in the **production** of goods and services (Shutters et al., 2018 PLoS One).

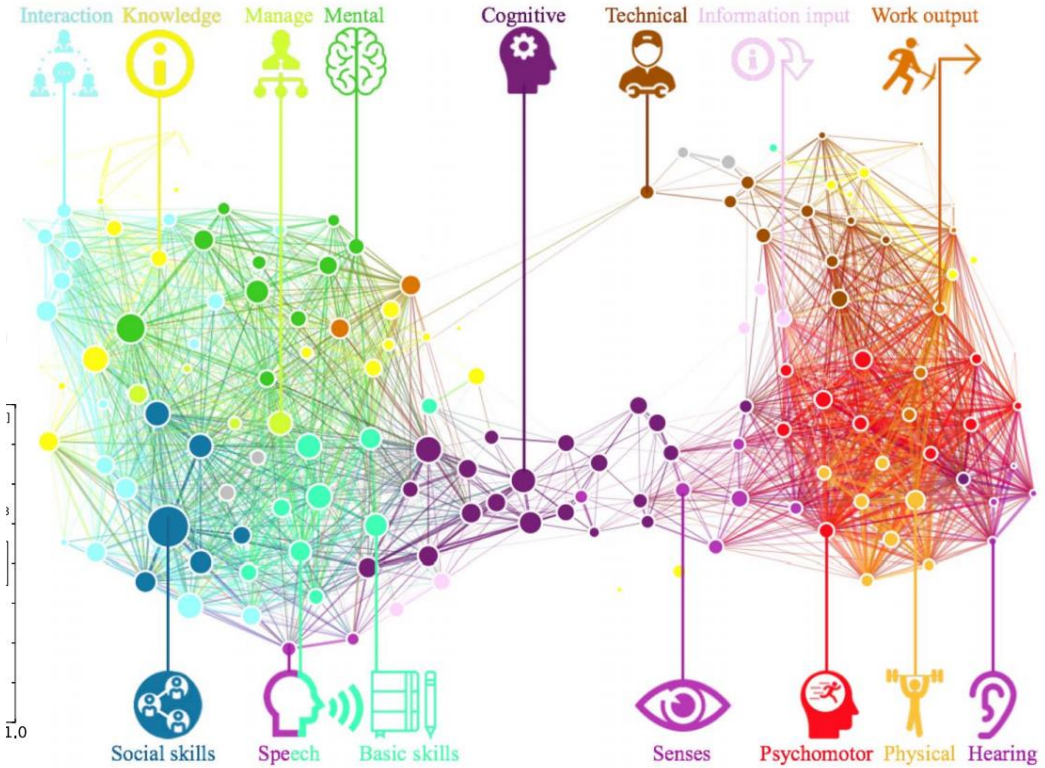
- **Relatedness**: overlap between capability requirements of activities.
  - Co-occurrence of technologies on patents (Tóth et al., 2022 EG).
  - Co-location of occupations (Shutters et al., 2018 PLoS One).
  - Normalised labour mobility (revealed skill relatedness) (Neffke et al., 2017 JEBO).
  - Similar skill demand (Alabdulkareem et al., 2018 SciAdv, Henning et al., 2025 RSRS).

Relatedness spaces

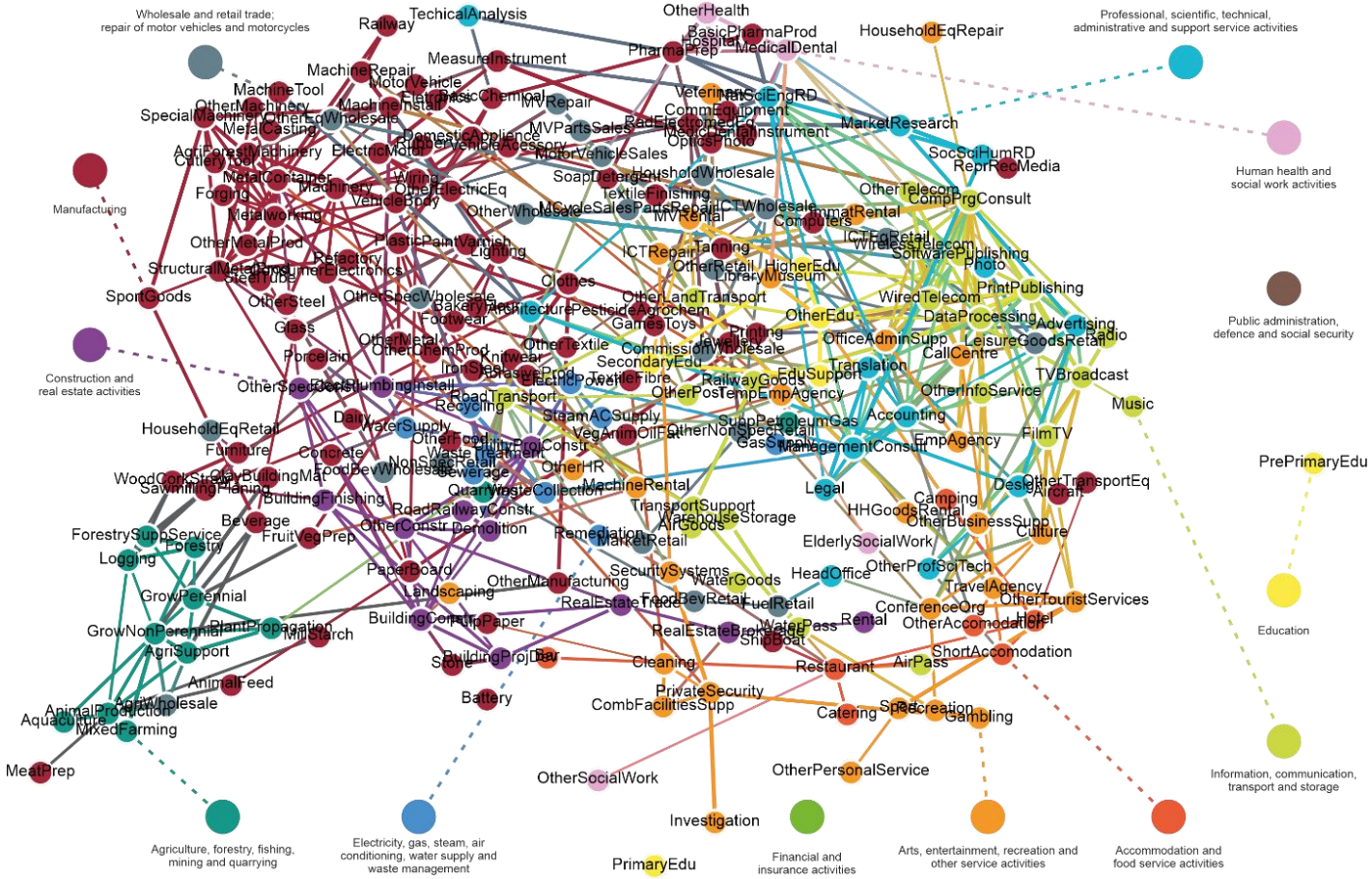
- Economic activities are related through similarities in skill usage (e.g., Alabdulkareem et al. 2018 SciAdv, Hosseinioun et al., 2025 NatHB).



Alabdulkareem et al. (2025) SciAdv



Relatedness spaces



Revealed skill relatedness (Neffke et al., 2017 JEBO).

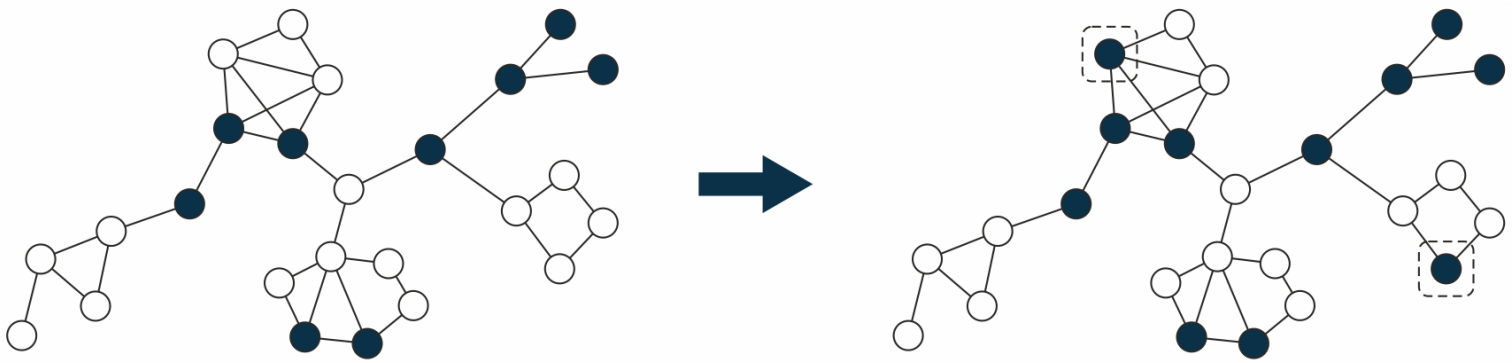
observed flow

$$SR_{ij} = \frac{F_{ij}}{\frac{F_{i.}F_{.j}}{F_{..}}}$$

expected flow

Principle of relatedness

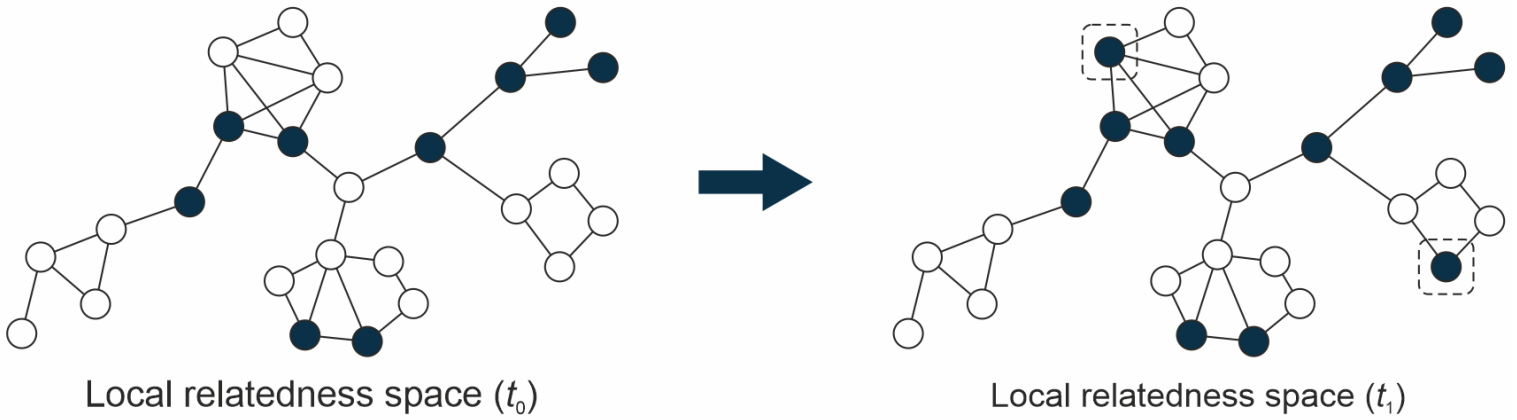
- New economic activities tend to enter places where there are related activities already present (e.g., Hidalgo et al. 2007 Science; Neffke et al., 2011 EG, Boschma et al., 2013 EG, Hidalgo et al., 2018).



NETFLIX
for places

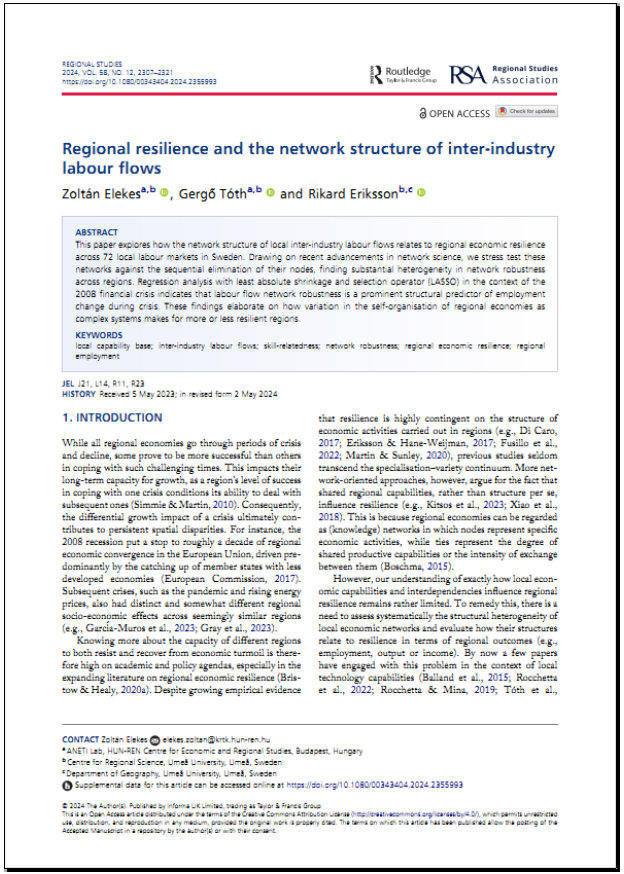
https://os-sweden.rec-group.se/

What I am after



- **Problem 1**: **local solutions** to the coordination of human capital through labour flows **may create more or less resilient local labour markets**.

- **Problem 2**: if productive capabilities are unevenly distributed across different workers, then **related diversification may align with some more than others.**



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Regional resilience and the network structure of inter-industry labour flows

Zoltán Elekes[a,b], Gergő Tóth[a,b] and Rikard Eriksson[b,c]

ABSTRACT
This paper explores how the network structure of local inter-industry labour flows relates to regional economic resilience across 72 local labour markets in Sweden. Drawing on recent advancements in network science, we stress test these networks against the sequential elimination of their nodes, finding substantial heterogeneity in network robustness across regions. Regression analysis with least absolute shrinkage and selection operator (LASSO) in the context of the 2008 financial crisis indicates that labour flow network robustness is a prominent structural predictor of employment change during crisis. These findings elaborate on how variation in the self-organisation of regional economies as complex systems makes for more or less resilient regions.

KEYWORDS
local capability base; inter-industry labour flows; skill-relatedness; network robustness; regional economic resilience; regional employment

JEL J21, L14, R11, R23
HISTORY Received 5 May 2023; in revised form 2 May 2024

1. INTRODUCTION

While all regional economies go through periods of crisis and decline, some prove to be more successful than others in coping with such challenging times. This impacts their long-term capacity for growth, as a region's level of success in coping with one crisis conditions its ability to deal with subsequent ones (Simmie & Martin, 2010). Consequently, the differential growth impact of a crisis ultimately contributes to persistent spatial disparities. For instance, the 2008 recession put a stop to roughly a decade of regional economic convergence in the European Union, driven predominantly by the catching up of member states with less developed economies (European Commission, 2017). Subsequent crises, such as the pandemic and rising energy prices, also had distinct and somewhat different regional socio-economic effects across seemingly similar regions (e.g., García-Muros et al., 2023; Gray et al., 2023).

Knowing more about the capacity of different regions to both resist and recover from economic turmoil is therefore high on academic and policy agendas, especially in the expanding literature on regional economic resilience (Bristow & Healy, 2020a). Despite growing empirical evidence that resilience is highly contingent on the structure of economic activities carried out in regions (e.g., Di Caro, 2017; Eriksson & Hane-Weijman, 2017; Fusillo et al., 2022; Martin & Sunley, 2020), previous studies seldom transcend the specialisation–variety continuum. More network-oriented approaches, however, argue for the fact that shared regional capabilities, rather than structure per se, influence resilience (e.g., Kitsos et al., 2023; Xiao et al., 2018). This is because regional economies can be regarded as (knowledge) networks in which nodes represent specific economic activities, while ties represent the degree of shared productive capabilities or the intensity of exchange between them (Boschma, 2015).

However, our understanding of exactly how local economic capabilities and interdependencies influence regional resilience remains rather limited. To remedy this, there is a need to assess systematically the structural heterogeneity of local economic networks and evaluate how their structures relate to resilience in terms of regional outcomes (e.g., employment, output or income). By now a few papers have engaged with this problem in the context of local technology capabilities (Balland et al., 2015; Rocchetta et al., 2022; Rocchetta & Mina, 2019; Tóth et al.,

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Regional resilience and the network structure of inter-industry labour flows



Zoltán Elekes



Gergő Tóth



Rikard Eriksson

Motivation

- In an evolutionary framework, resilience is the capacity of regional economies to withstand economic shocks and at the same time to retain their long-term ability to develop new growth paths (Boschma, 2015 RS).

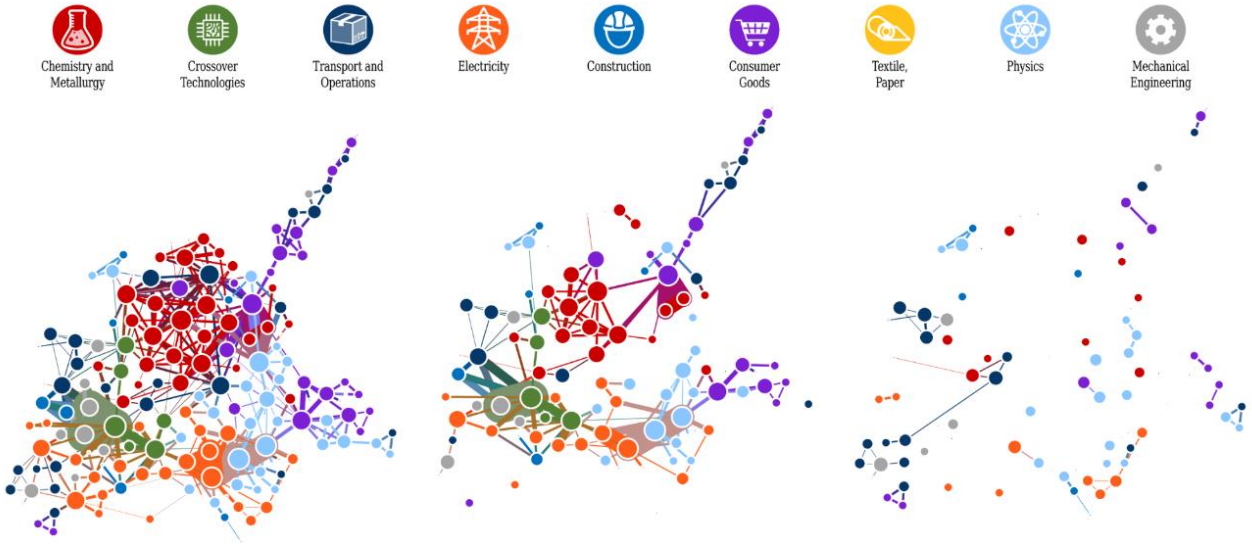




- "[…] in the regional resilience literature, it is remarkable **how little attention has been paid to the sensitivity of regional networks to the removal of specific nodes or the dissolution of particular linkages**. One can depict a regional economy as a knowledge network in which the nodes stand for **industries/technologies** and the ties reflect the degree of technological relatedness between these nodes" Boschma (2015 RS, p. 714).

Motivation

- Embeddedness in local technology space predicts inventive performance during crisis (Balland et al. 2015, CJRES).
- Embeddedness in local occupation space predicts unemployment during crisis (Moro et al. 2021, NatComm).
- The robustness of local technology networks predicts employment during crisis (Tóth et al. 2022, EG).



Research design

- Matched employer-employee data from Swedish registers aggregated to 3-digit industry codes in 72 functional labour market regions.

- Revealed skill relatedness networks of industries based on Neffke et al., (2017) JEBO.

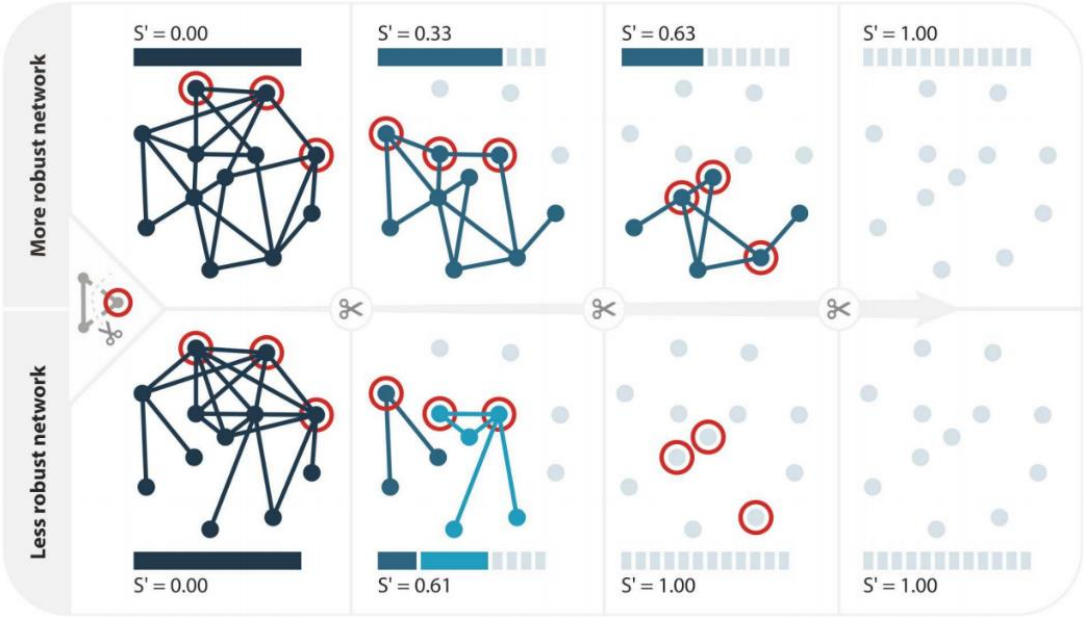
- Identifying significant labour flows between industries from 2002-2007.

$$SR_{ij} = \frac{F_{ij}}{\frac{F_{i.}F_{.j}}{F_{..}}}$$

observed flows

expected (local) flows

Research design



- Entropy-based network robustness measure (Zitnik et al., 2019 PNAS).

$$S'(G_i^f) = S(G_i^f)/\log N$$

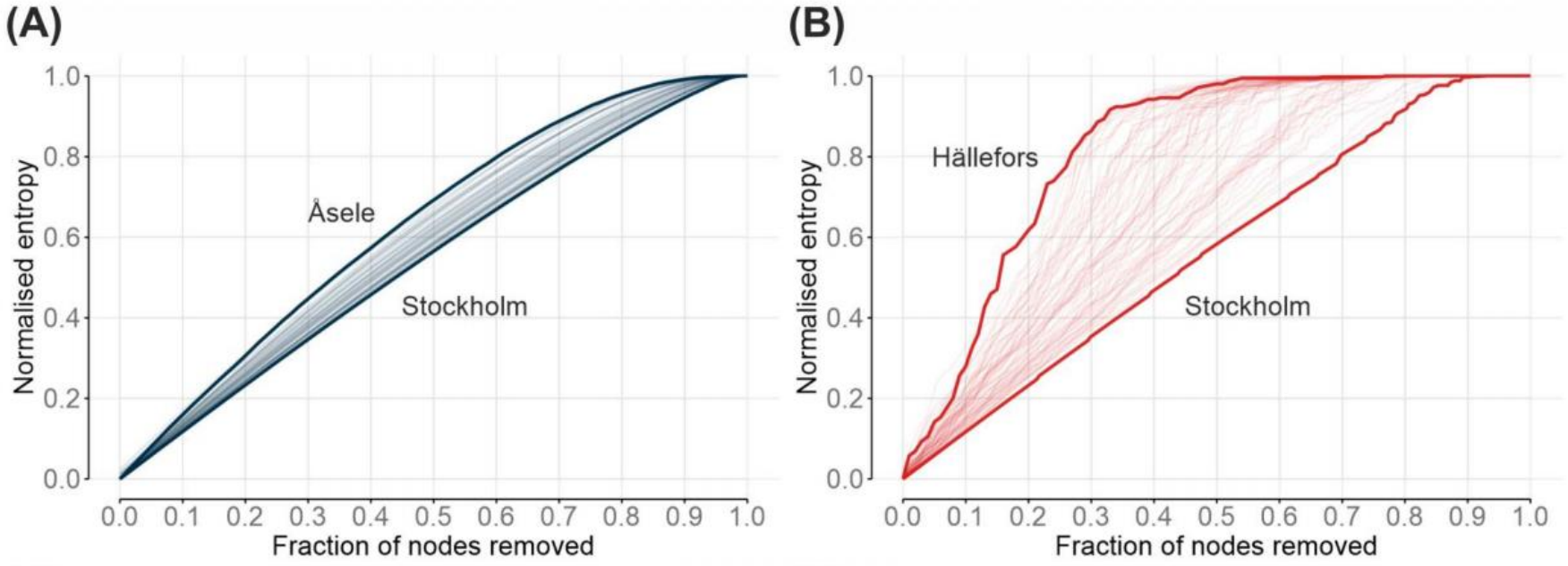
- Sequential elimination of nodes (industries) from local networks: random and targeted.

- Network robustness (Ω) measures how quickly a network fragments to disconnected components.

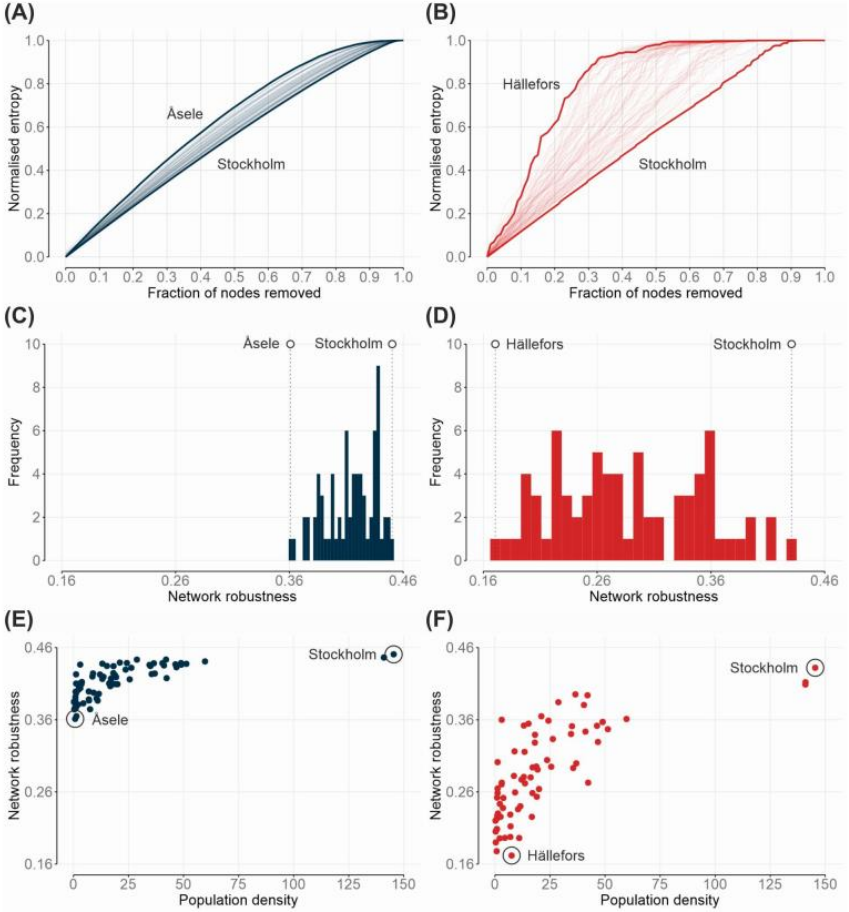
Results

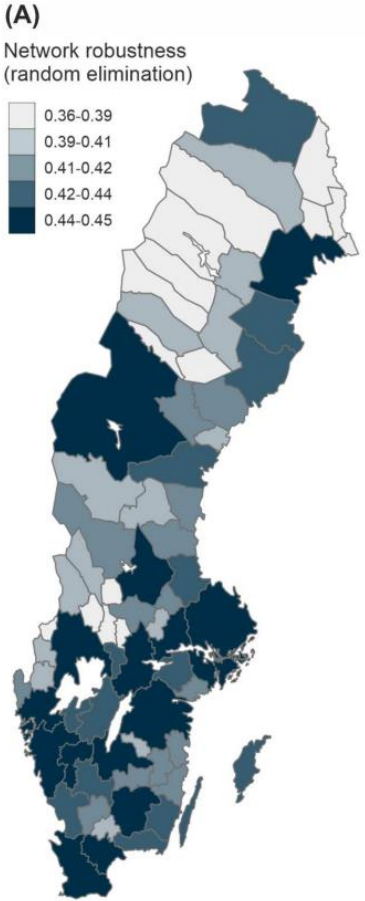
Random removal

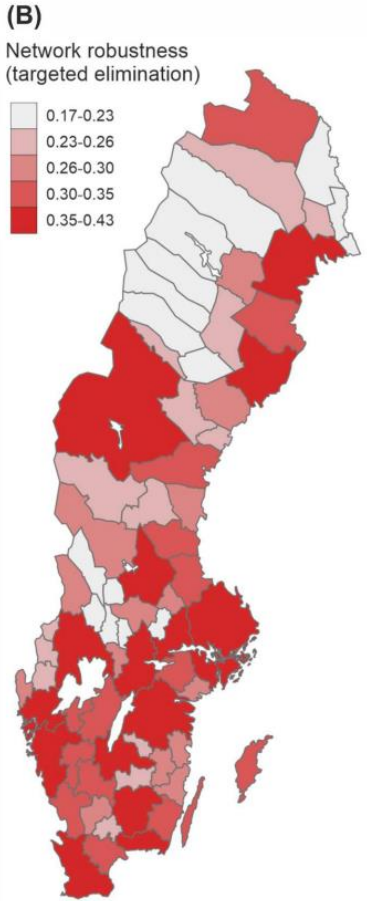
Targeted removal



Results


(A)
Normalised entropy
Åsele
Stockholm
Fraction of nodes removed
(B)
Normalised entropy
Hällefors
Stockholm
Fraction of nodes removed
(C)
Åsele
Stockholm
Frequency
Network robustness
(D)
Hällefors
Stockholm
Frequency
Network robustness
(E)
Stockholm
Åsele
Network robustness
Population density
(F)
Stockholm
Hällefors
Network robustness
Population density

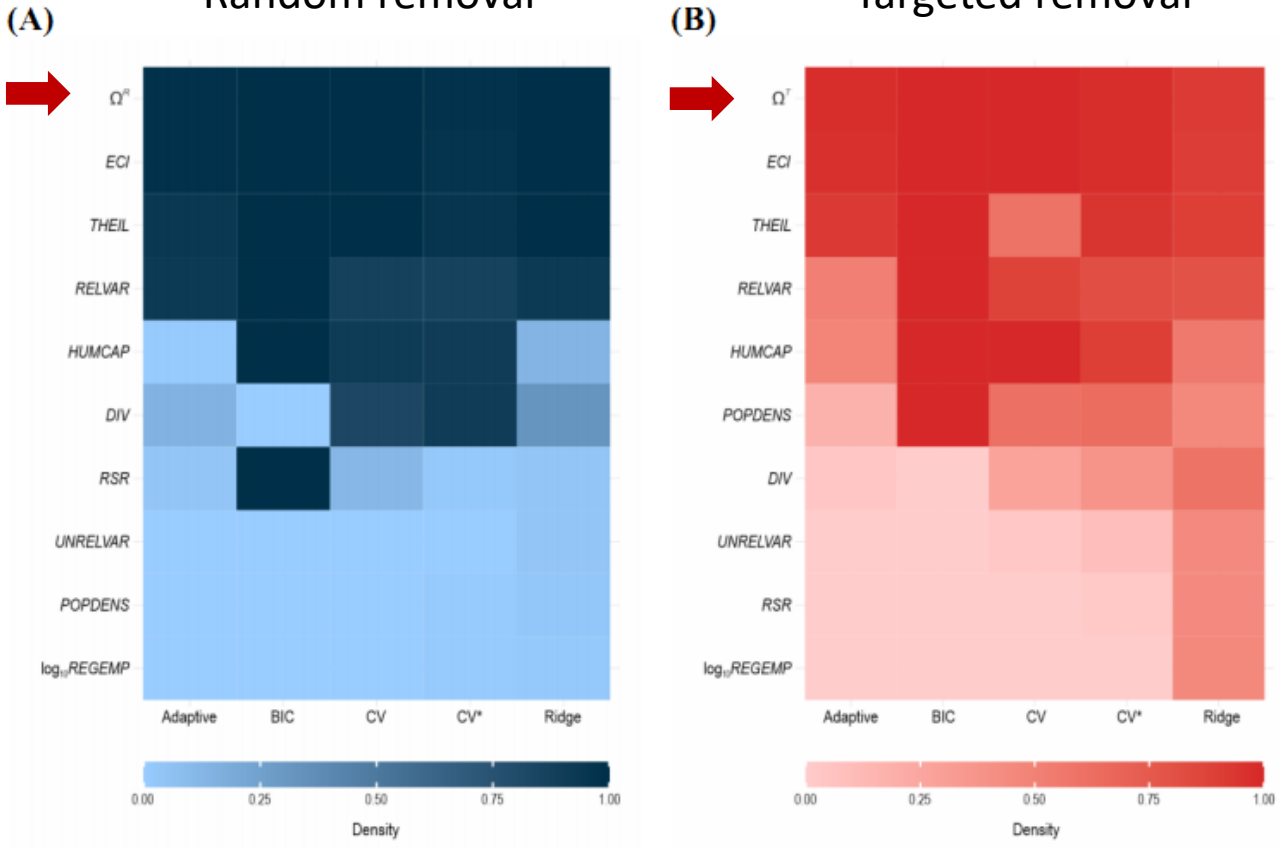

(A)
Network robustness
(random elimination)
0.36-0.39
0.39-0.41
0.41-0.42
0.42-0.44
0.44-0.45


(B)
Network robustness
(targeted elimination)
0.17-0.23
0.23-0.26
0.26-0.30
0.30-0.35
0.35-0.43

Results

Random removal

Targeted removal



- **Predicting employment change** with structural variables, including network robustness.
- **Least absolute shrinkage and selection operator (LASSO)** inference and variable selection.

Conclusions

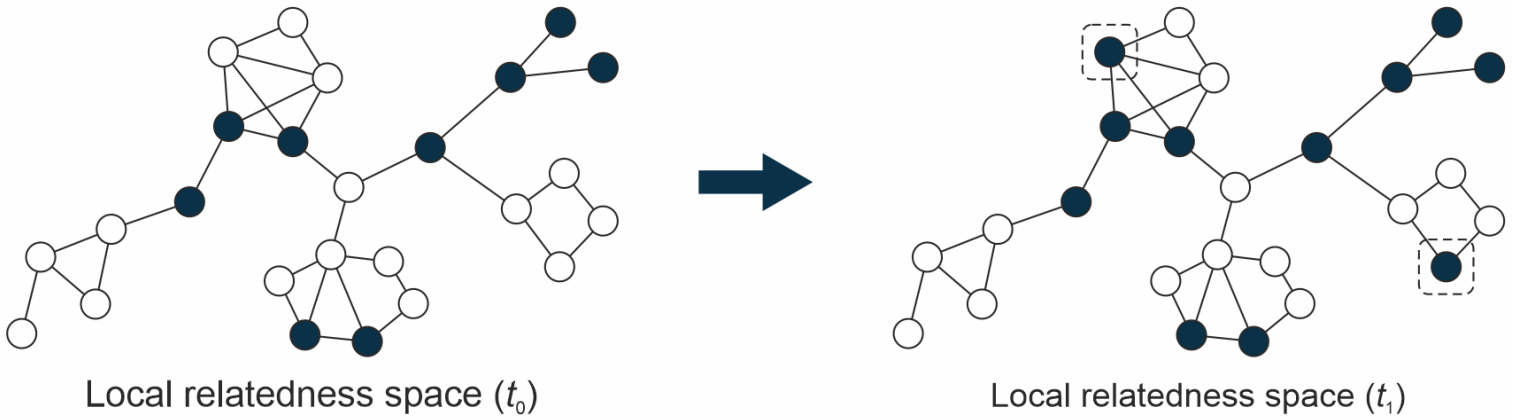
(1) Particular solutions to local skill coordination problems via **labour redeployment flows** show **heterogeneous capacity** to withstand disturbances.

(2) **Validating** the role labour flow **network robustness** in the context of a **grand recession**.

(3) Findings go **beyond urban areas** and show heterogeneity in robustness among smaller regions.

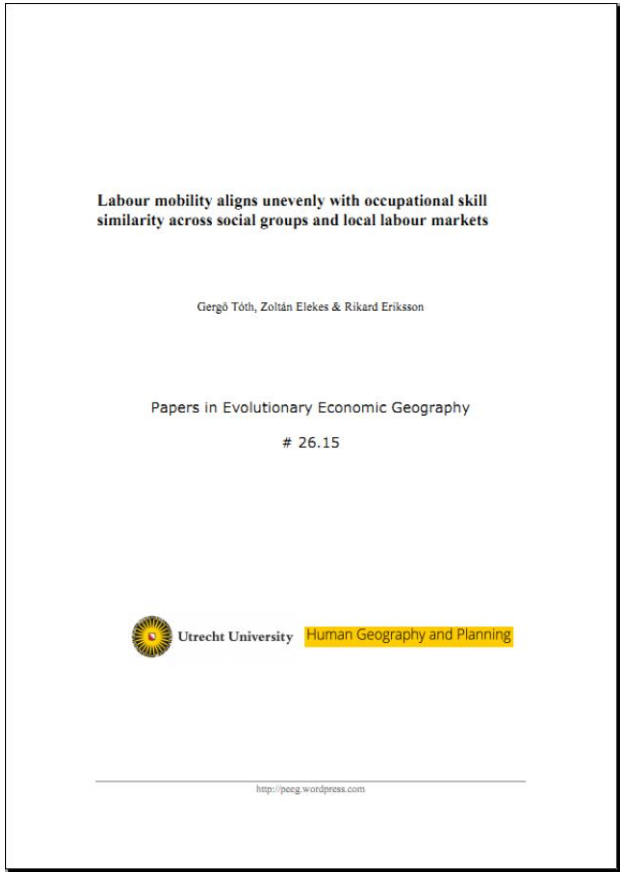
(4) Network robustness is a **prominent predictor** of employment change among established structural measures, thereby showing there is **more to local economic structure** than the **distribution of workers** across activities, or **relatedness measured** from **national aggregates**.

What I am after



- **Problem 1**: **local solutions** to the coordination of human capital through labour flows **may create more or less resilient local labour markets**.

- **Problem 2**: if productive capabilities are unevenly distributed across different workers, then **related diversification may align with some more than others.**



Labour mobility aligns unevenly with occupational skill similarity across social groups and local labour markets



Gergő Tóth



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Motivation

- Labour mobility is central to the productive redeployment of worker skills (Glaeser, 2005 JoEG; Boschma & Wenting, 2007 ICC; Neffke & Henning, 2013 SMJ; Neffke et al., 2017 JEBO).
- Labour markets do not allocate workers solely according to transferable skills and capabilities (Granovetter, 1995; Cardoso et al., 2025 NatHB).

- **(A)** If labour mobility primarily reflects efficient skill redeployment, workers should disproportionately move between occupations with **similar skill requirements**.

- **(B)** If, however, mobility systematically deviates from these pathways for particular social groups, this would suggest that labour market **segmentation structures opportunity** beyond differences in skill requirements.

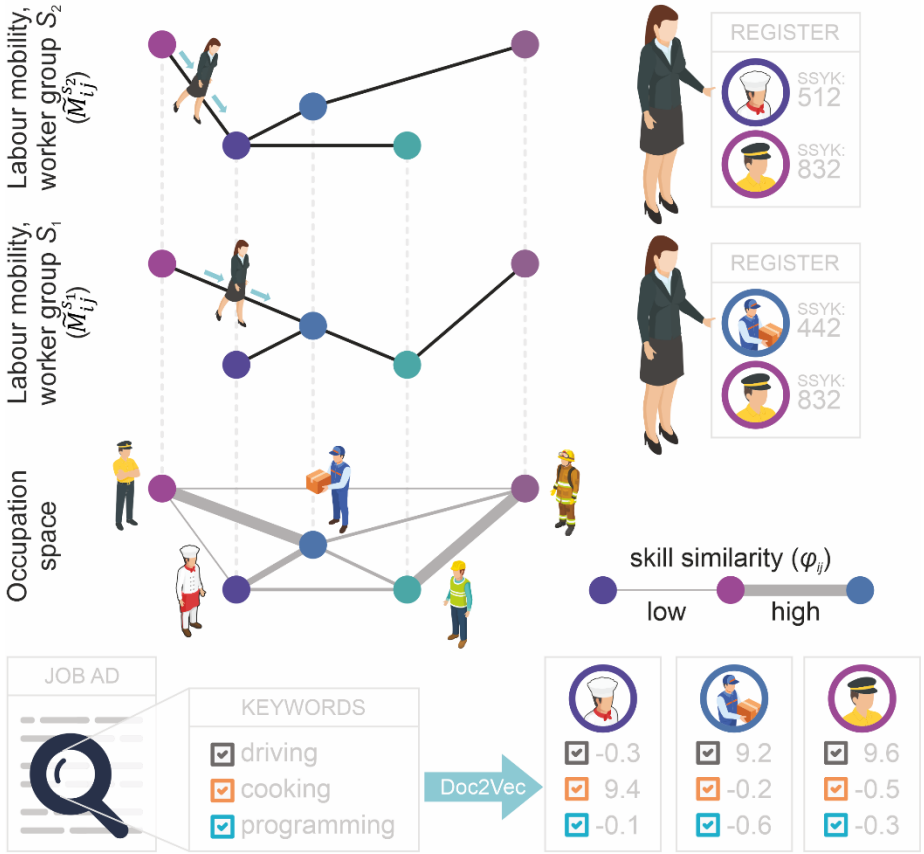
- Existing approaches offer partial solutions.

Methods

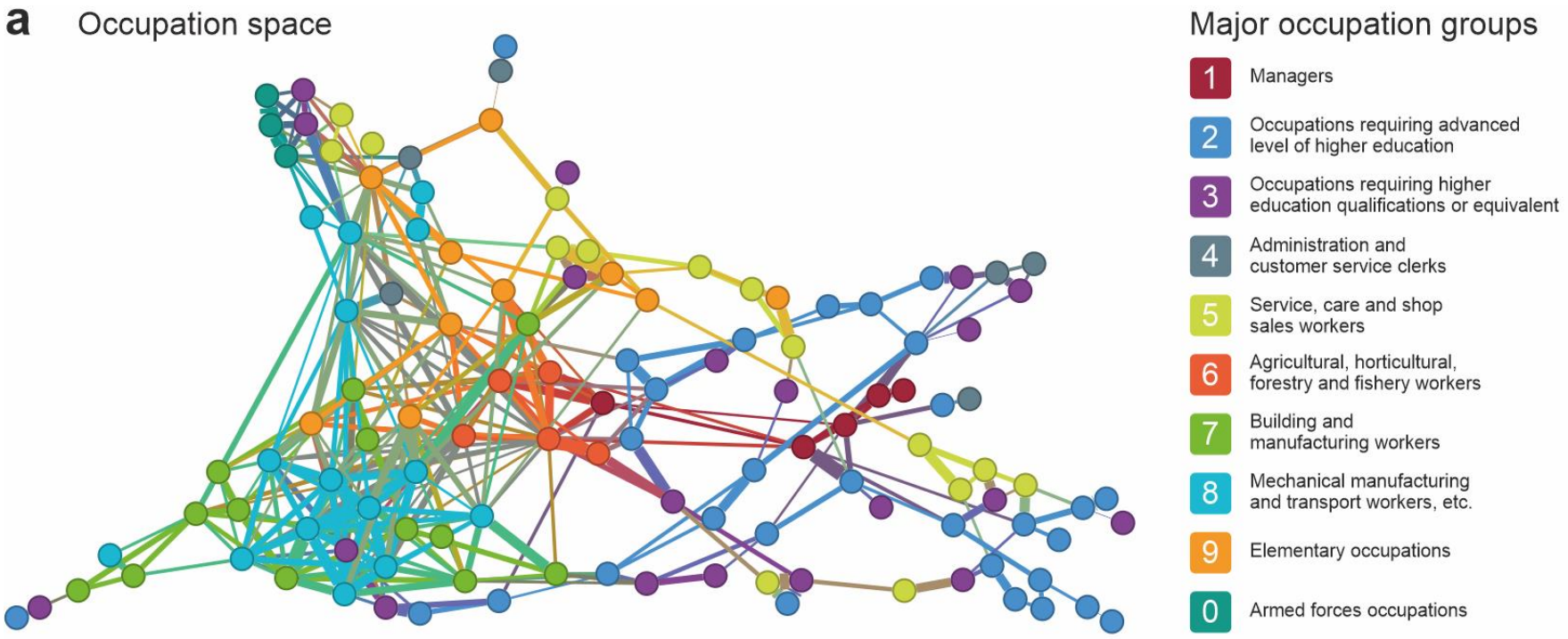
- **Occupations:** 122* 3-digit SSYK12 codes consistent with ISCO-08.
- **Regions:** 72 funcional labour markets.

- **Register microdata** from Statistics Sweden, including work history and demographical features
- **Labour mobility:** annual mobility of workers between occupations over 2016-2019.

- **700K job advertisments** for 2016 from the Swedish Public Employment Office (Arbetsförmedlingen)
- **Skill relatedness:** cosine similarity of skills demanded in pairs of occupations.



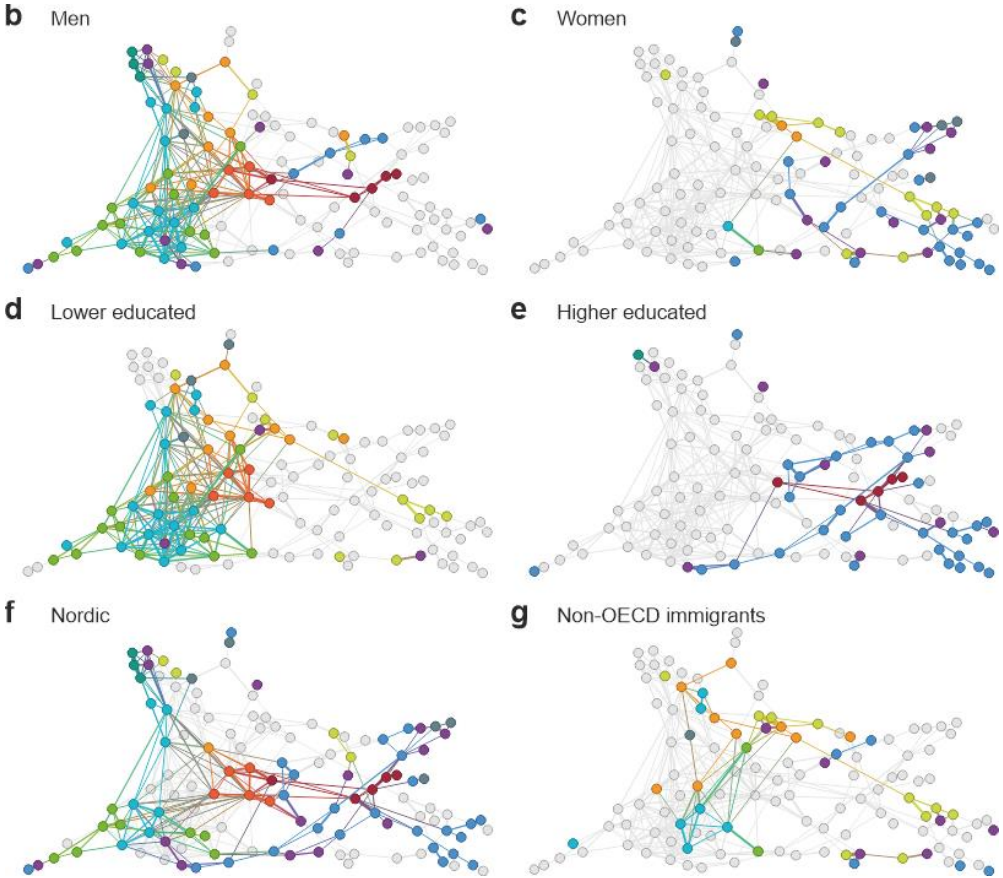
Results: the occupation space



Results: occupational segregation

- **Social groups** exhibit different relative specialisations in the occupation space:
  - Men and women.
  - Lower and higher educated.
  - Workers with Nordic and non-OECD immigrant background.

- Occupation space reflects the **occupational segregation** of social groups.



Results: skill relatedness explains labour mobility

Table 1. Skill similarity explains labour mobility.

	(1) OLS	(2) OLS	(3) OLS FE	(4) 2SLS	(5) LPM
φ_{ij}	0.508***	0.343***	0.631***	1.053***	0.134***
	(0.008)	(0.008)	(0.009)	(0.035)	(0.003)
θ_{ij}^{g}		0.204***	0.174***	0.059***	0.062***
		(0.006)	(0.007)	(0.011)	(0.003)
θ_{ij}^{e}		0.328***	0.306***	0.107***	0.092***
		(0.007)	(0.009)	(0.014)	(0.003)
θ_{ij}^{o}		0.100***	0.161***	0.132***	0.033***
		(0.006)	(0.011)	(0.009)	(0.003)
Constant	-0.000	-0.000	-0.107	0.041***	0.245***
	(0.007)	(0.006)	(0.107)	(0.009)	(0.003)
Occ. FE	No	No	Yes	No	No
N	12 656	12 656	12 656	11 772	12 656
R^2	0.258	0.413	0.592	0.049	0.249

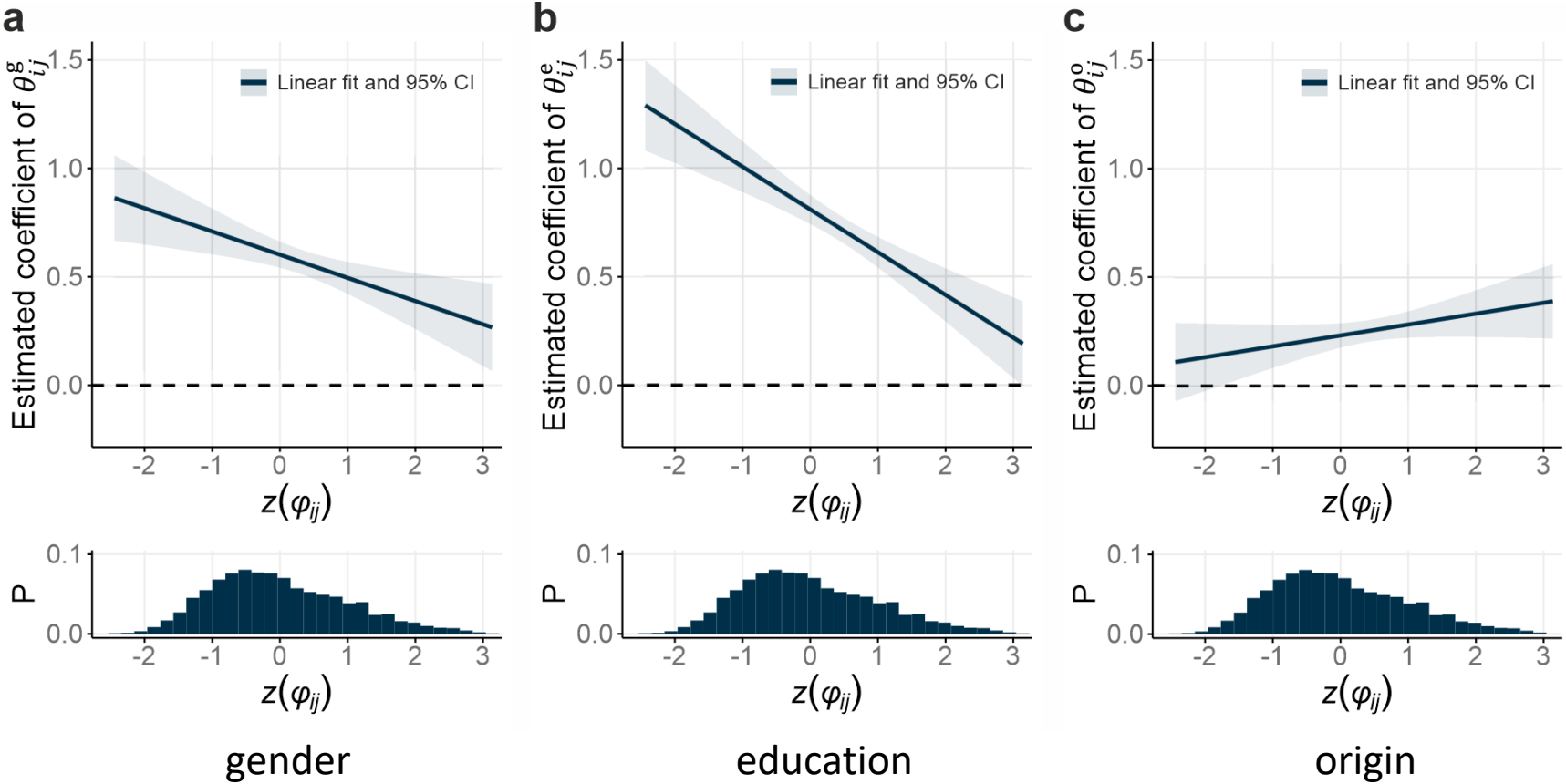
Notes: ***p < 0.001, **p < 0.01, *p < 0.05. Variables are standardised into z-scores (see equation 10).

- Explaining labour mobility with skill relatedness and demographic similarity.

$$\widetilde{M}_{ij} = \alpha + \beta\varphi_{ij} + \gamma_1\theta_{ij}^{g} + \gamma_2\theta_{ij}^{e} + \gamma_3\theta_{ij}^{o} + \zeta_i + \zeta_j + \varepsilon_{ij}$$

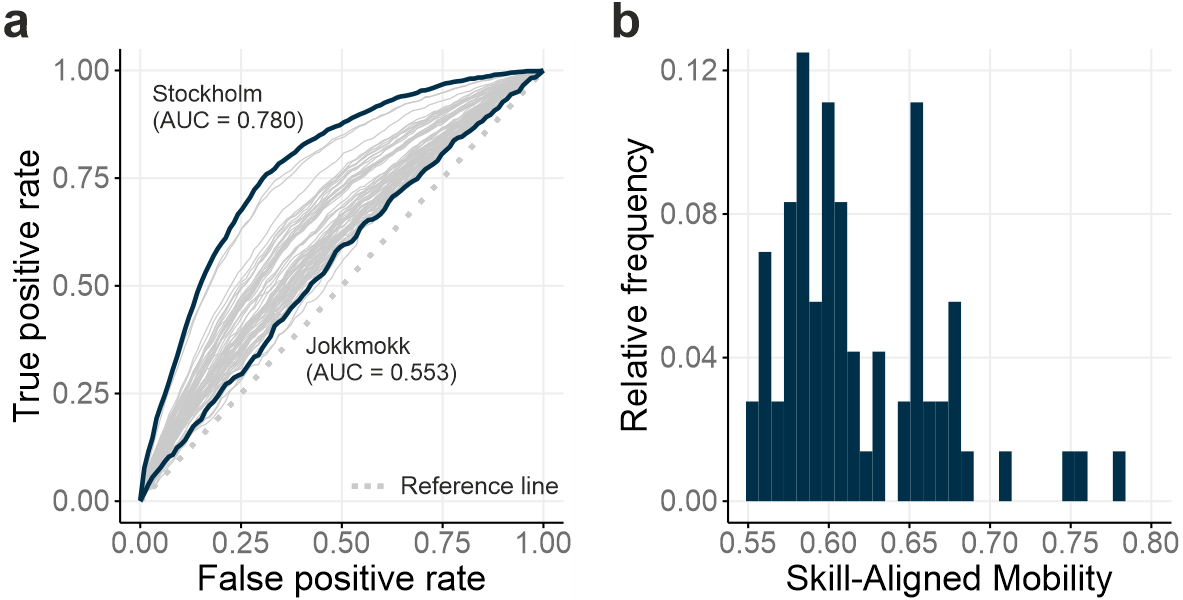
- Consistent positive association.
- Skill relatedness explains labour mobility. Instrumental variable: skill relatedness in O*NET.

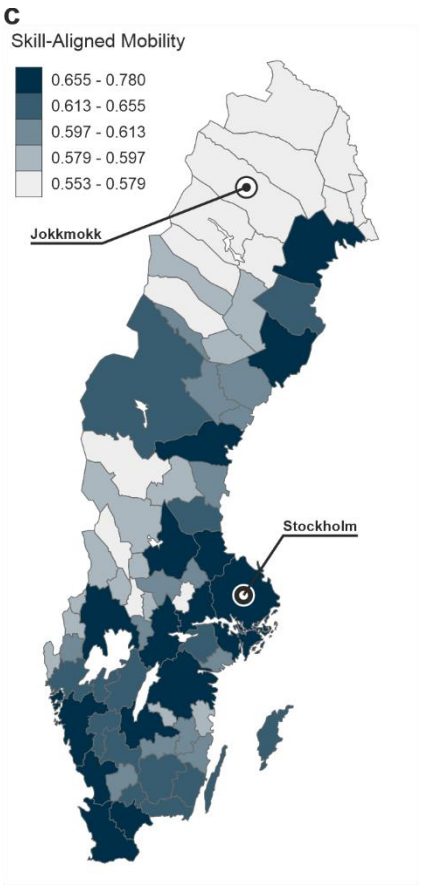
Results: moderating role of skill relatedness



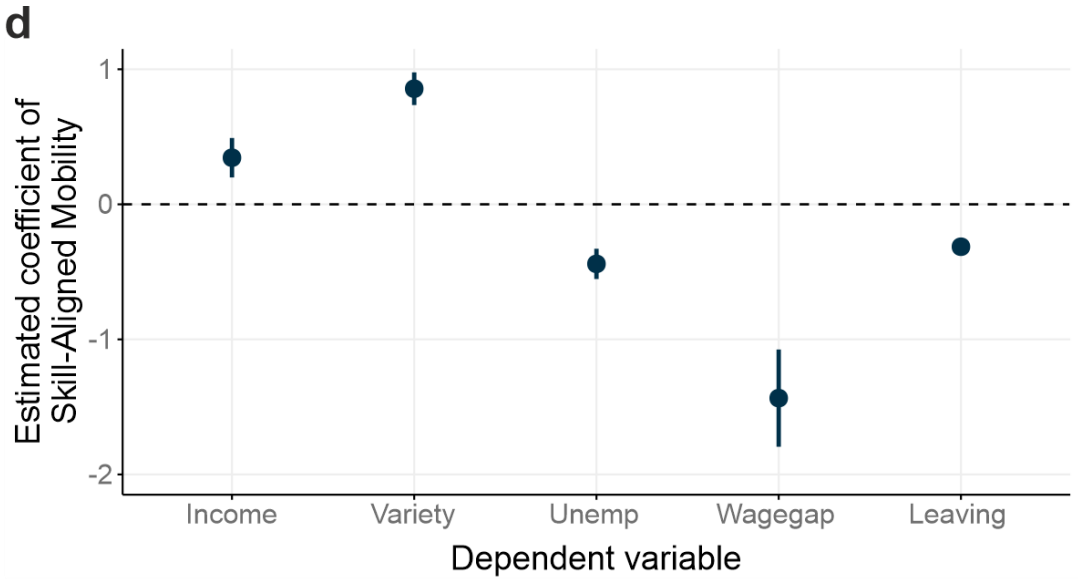
Results: outcomes on local labour markets

- **Skill-aligned mobility (SAM)**: to what extent does skill relatedness predict labour mobility in the local labour market (ROC-AUC for local labour markets).





Results: outcomes on local labour markets



	(1)	(2)	(3)	(4)	(5)
N	72	72	72	72	72
R^2	0.287	0.641	0.236	0.227	0.485

$$Y_R = \alpha + \beta \ln(SAM_R) + \gamma \, \text{Popdens}_R + \varepsilon_R$$

- Local labour markets with more skill-aligned mobility exhibit performance across the board.

Conclusion

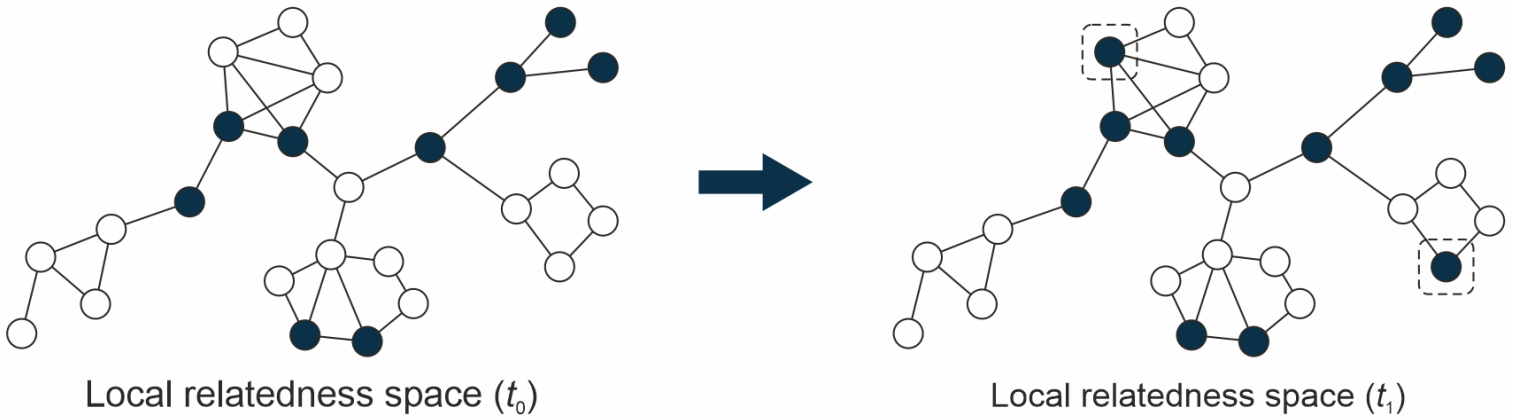
- (1) **Analytically separating** skill relatedness and significant labour flows to quantify how closely the two **align** across social groups and local labour markets.

- (2) Overall, labour mobility **follows skill-related pathways**. Instrumental variable estimates provide evidence **consistent with a causal effect** of skill relatedness on labour mobility.

- (3) **Access** to these pathways **is uneven**: occupational segregation along gender, education and country of birth shapes labour mobility beyond skill relatedness.

- (4) **Local labour markets** where mobility follows skill relatedness more closely exhibit more favourable labour-market outcomes.

Summary



- **Problem 1**: **local solutions** to the coordination of human capital through labour flows **may create more or less resilient local labour markets**.

- **Problem 2**: if productive capabilities are unevenly distributed across different workers, then **related diversification may align with some more than others.**

Thank you!



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